STAT 412

Module 9 Template File for Final Project

Graded at 100 points

NAME: ______________________________________

	Analysis to be Performed	**EXCEL COMMAND USED (if applicable)**	**NUMERIC RESULT or text- based response**
Section 1 **<u>Confidence Interval for Mean</u>** (15 points total, 5 points each part)	a) Based on your random sample of 250 rows, formulate a point estimate for the mean arrival time		
	b) Based on your random sample of 250 rows, generate a 95% confidence interval for the mean arrival time delay. Be sure to include any EXCEL commands and supporting calculations.		
	c) Formulate a one-sentence interpretation of this confidence interval. Imagine you are explaining the meaning of this confidence interval to someone that does not have a statistical background.		

Section 2 **Confidence Interval for Proportion** (15 points total, 5 points each part)	a) Based on your random sample of 250 rows, formulate a point estimate for the proportion of late arrivals.		
	b) Based on your random sample of 250 rows, generate a 95% confidence interval for the proportion of late arrivals. Be sure to include any EXCEL commands and supporting calculations.		
	c) Formulate a one-sentence interpretation of this confidence interval. Imagine you are explaining the meaning of this confidence interval to someone that does not have a statistical background.		
Section 3 **Hypothesis Test for Mean Based on One Sample** (20 points total, 4 points each part)	a) Set up the null and alternative hypothesis. Be sure to set this up using H_0 and H_1 notation and use correct symbols for the population parameter of interest.		
	b) Calculate the test statistic. Include any Excel commands and supporting calculations.		

	c) Calculate the corresponding p-value. Include any Excel commands and supporting calculations.		
	d) Clearly state your decision as to whether you “reject the null hypothesis” or “do not reject the null hypothesis”. Explain the rationale for your decision.		
	e) Compose an email to the manager explaining your findings. Describe your test methodology and formulate a concluding statement regarding the managers claim. Do you support the managers claim, or not support the claim? Explain the rationale for your decision in language that the manager can understand and assume the manager does not have a statistical background.		

Section 4 **<u>Hypothesis Test for Proportion Based on One Sample</u>** <u>(20 points total, 4 points each part)</u>	a) Set up the null and alternative hypothesis. Be sure to set this up using H0 and H1 notation and use correct symbols for the population parameter of interest.		
	b) Calculate the test statistic. Include any Excel commands and supporting calculations.		
	c) Calculate the corresponding p-value. Include any Excel commands and supporting calculations.		
	d) Clearly state your decision as to whether you “reject the null hypothesis” or “do not reject the null hypothesis”. Explain the rationale for your decision.		
	e) Compose an email to the manager explaining your findings. Describe your test methodology and formulate a concluding statement regarding the managers claim. Do you support the managers claim, or		

	not support the claim? Explain the rationale for your decision in language that the manager can understand and assume the manager does not have a statistical background.		
Section 5 **<u>Hypothesis Test for Mean Based on Two Samples</u>** <u>(30 points total, 6 points each part)</u>	a) Set up the null and alternative hypothesis. Be sure to set this up using H0 and H1 notation and use correct symbols for the population parameter of interest.		
	b) Calculate the test statistic. Include any Excel commands and supporting calculations.		
	c) Calculate the corresponding p-value. Include any Excel commands and supporting calculations.		
	d) Clearly state your decision as to whether you "reject the null hypothesis" or "do not reject the null hypothesis". Explain the rationale for your decision.		
	e) Compose an email to the CEO of your airline explaining your findings. Describe your test		

	methodology and formulate a concluding statement regarding whether the mean arrival time delay for your airline is better that the mean arrival delay time for a competitor airline. Explain the rationale for your decision in language that the CEO can understand and assume the CEO does not have a statistical background.		